

CSE440: Natural Language Processing II

Dr. Farig Sadeque
Associate Professor
Department of Computer Science and Engineering
BRAC University

Lecture 6: Recurrent Neural Nets

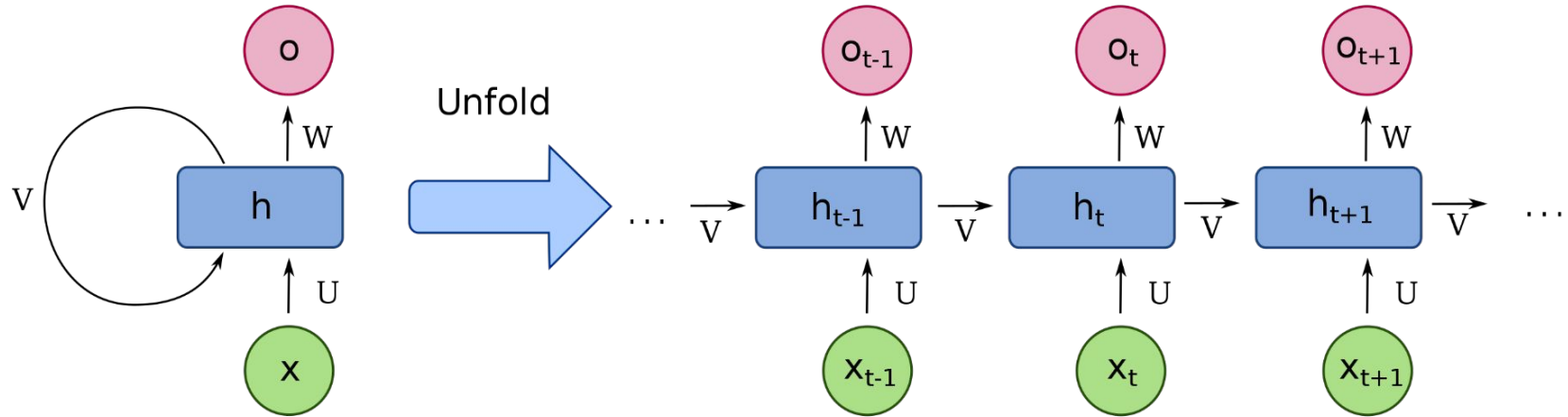
Recurrent Neural Networks (RNN)

- Simple recurrent networks
- Bidirectional and gated recurrent networks
- Recurrent architectures
- Seq2Seq models (next session)
- Attention (next session)

Short history of RNN

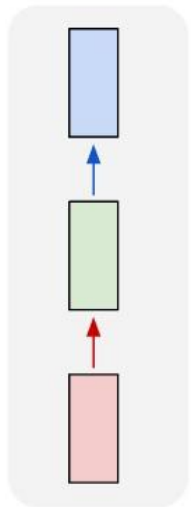
- 1986: RNNs are Introduced by David Rumelhart
- 1995: LSTMs are introduced by Sepp Hochreiter and Jürgen Schmidhuber based on Hochreiter's 1991 research on vanishing gradient problem
- 2001: Gers and Schmidhuber trained LSTMs to learn language models (unlearnable by HMMs)
- 2009: Graves et al. won ICDAR handwriting recognition competition using LSTMs
- 2013: Hinton and his team destroyed previous record for speech recognition using LSTM
- 2014: GRU is introduced by Cho et al.
- 2015: Widespread use in both academia and industry due to Google's adaptation of LSTM in their Google Voice speech recognition system

Structure of an RNN

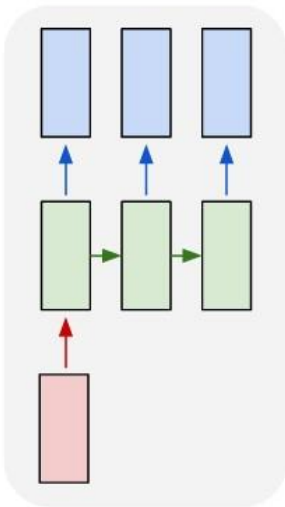


Types of RNNs

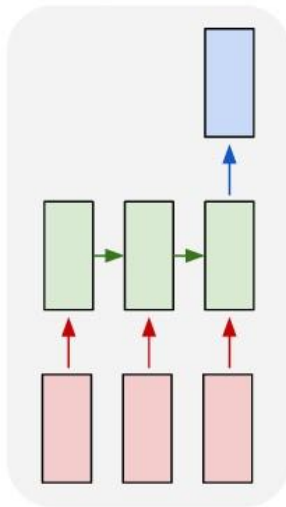
one to one



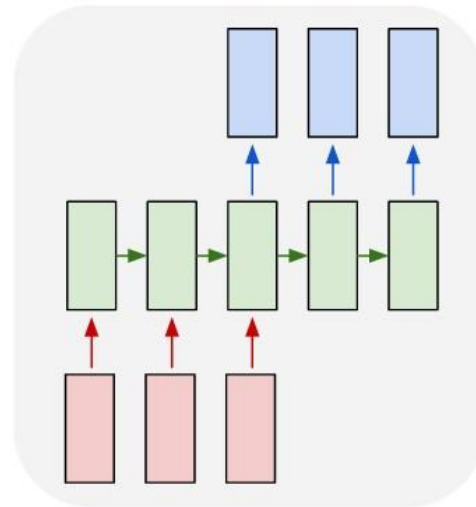
one to many



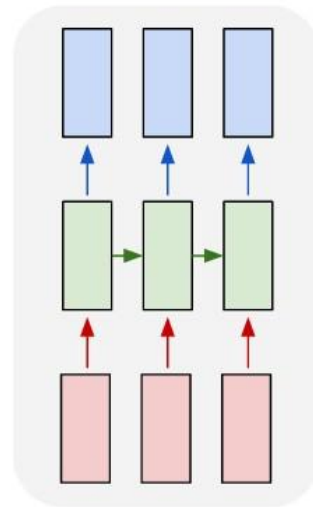
many to one



many to many

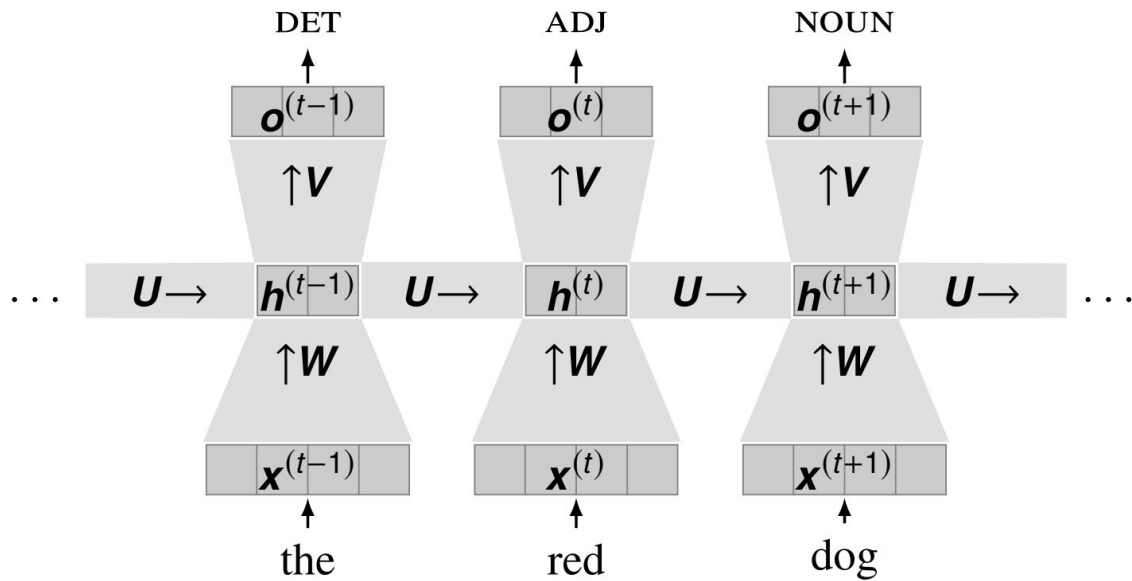


many to many



<https://karpathy.github.io/2015/05/21/rnn-effectiveness/>

A completely unrolled Simple RNN



$$\mathbf{o}^{(t)} = \text{softmax}(\mathbf{V}\mathbf{h}^{(t)})$$

$$\mathbf{h}^{(t)} = \tanh(\mathbf{U}\mathbf{h}^{(t-1)} + \mathbf{W}\mathbf{x}^{(t)})$$

Simple RNN

Intuitions:

- Each step combines the current input with the history
- Each prediction is made based on this combination

Observations:

- The input and hidden state change at each time step
- The parameters W , U , V are the same at each step

Equations:

$$\mathbf{h}^{(t)} = \tanh(\mathbf{U}\mathbf{h}^{(t-1)} + \mathbf{W}\mathbf{x}^{(t)})$$

$$\mathbf{o}^{(t)} = \text{softmax}(\mathbf{V}\mathbf{h}^{(t)})$$

Classwork

Consider an RNN that predicts as:

$$\mathbf{o}^{(t)} = \text{softmax}(\mathbf{V}\mathbf{h}^{(t)})$$

$$\mathbf{h}^{(t)} = \mathbf{U}\mathbf{h}^{(t-1)} + \mathbf{W}\mathbf{x}^{(t)}$$

whose parameters have been set to:

$$\mathbf{U} = \begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix} \quad \mathbf{W} = \begin{bmatrix} 2 & 0 & -1 \\ 1 & -1 & 1 \end{bmatrix} \quad \mathbf{V} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ 0 & 3 \end{bmatrix}$$

If you are given the following input:

$$\mathbf{h}^{(0)} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad \mathbf{x}^{(1)} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \mathbf{x}^{(2)} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Which label will be predicted for each word if in the final softmax, index 0=ADJ, index 1=DET, and index 2=NOUN?

Drawbacks of simple RNN

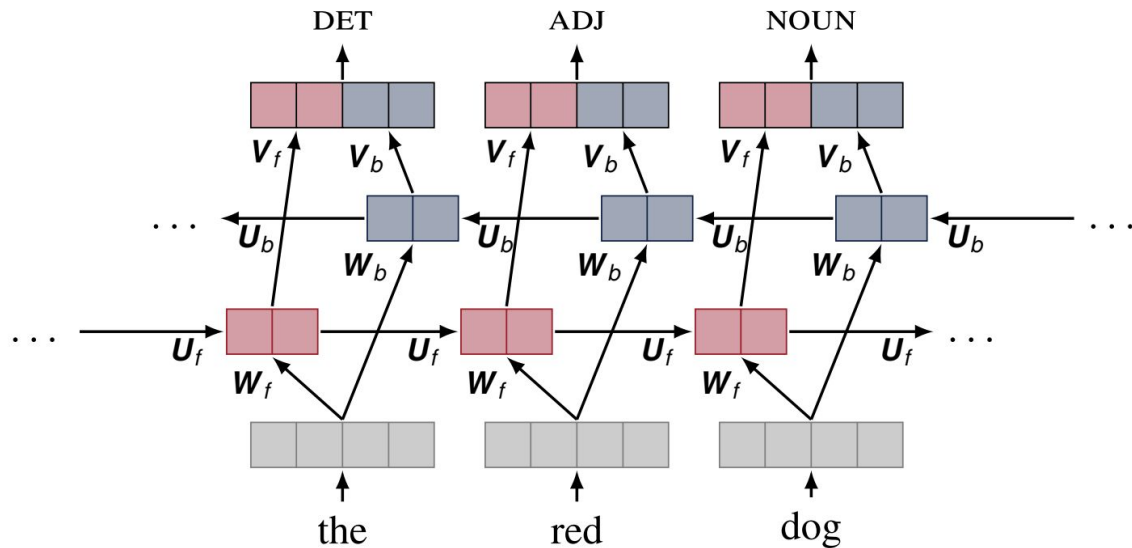
- They can only see the past, not the future
- They must forget the same amount of history at each time step
 - Theoretically, they don't have to
 - Maintaining long term memory is difficult

Bidirectional RNNs

Intuition:

- Run one forward RNN
- Run one backward RNN
- Combine their outputs

Bidirectional RNNs



$$\mathbf{o}^{(t)} = \text{softmax}(\mathbf{V}_f \mathbf{h}_f^{(t)} + \mathbf{V}_b \mathbf{h}_b^{(t)})$$

$$\mathbf{h}_b^{(t)} = \tanh(\mathbf{U}_b \mathbf{h}^{(t+1)} + \mathbf{W}_b \mathbf{x}^{(t)})$$

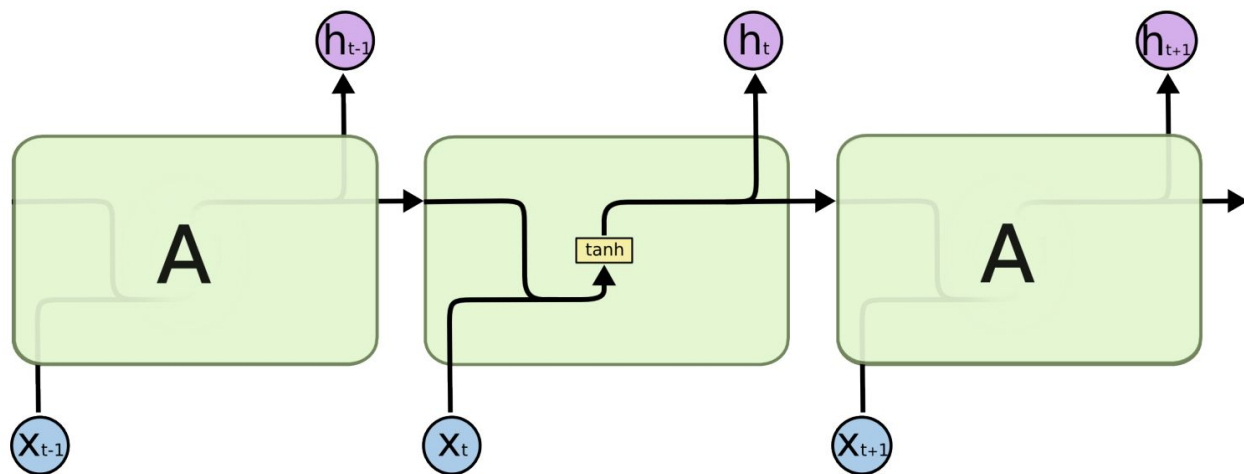
$$\mathbf{h}_f^{(t)} = \tanh(\mathbf{U}_f \mathbf{h}^{(t-1)} + \mathbf{W}_f \mathbf{x}^{(t)})$$

Gated RNNs

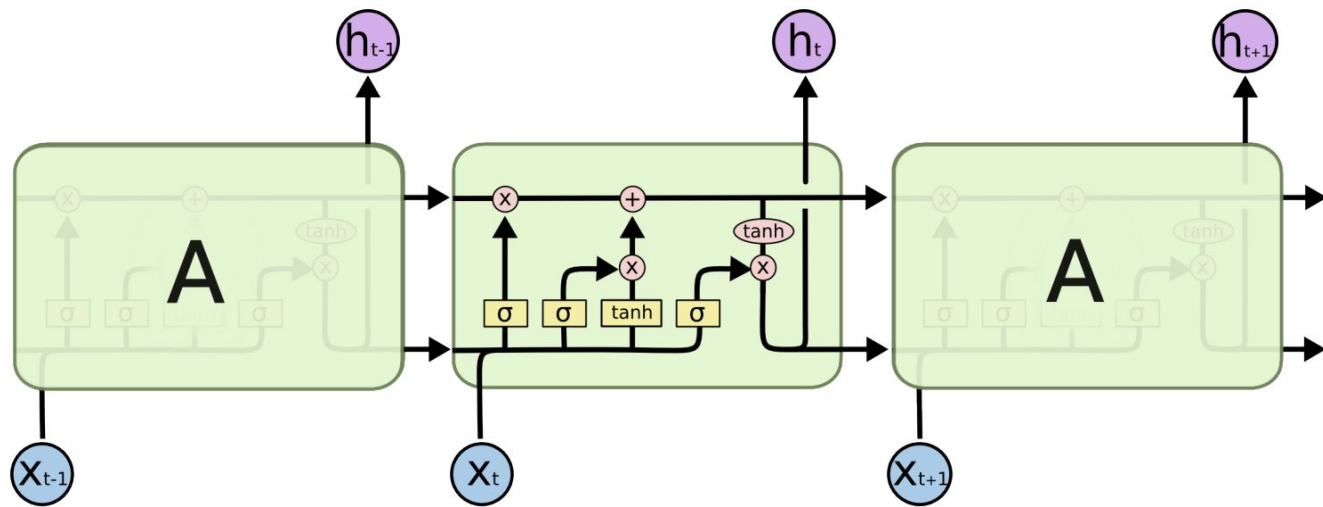
Intuition:

- Simple RNNs forget the same amount at each time step
- Look at the previous hidden state and the current input
- Decide how much to forget based on those
- Two gated RNNs
 - Long Short-Term Memory (LSTM)
 - Gated Recurrent Units (GRU)

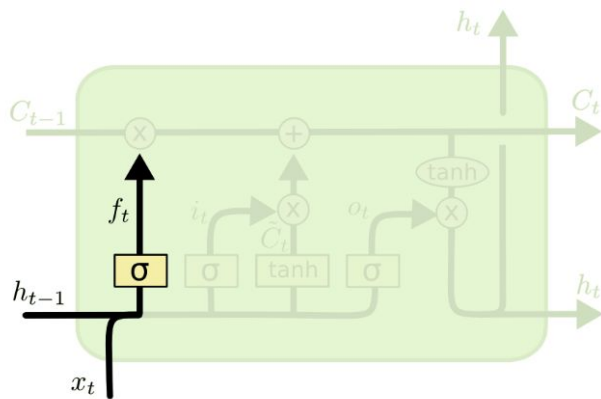
Simple RNN



LSTM

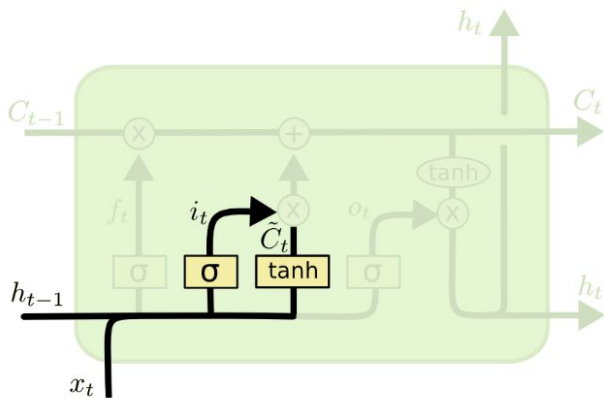


Forget gate



$$f_t = \sigma (W_f \cdot [h_{t-1}, x_t] + b_f)$$

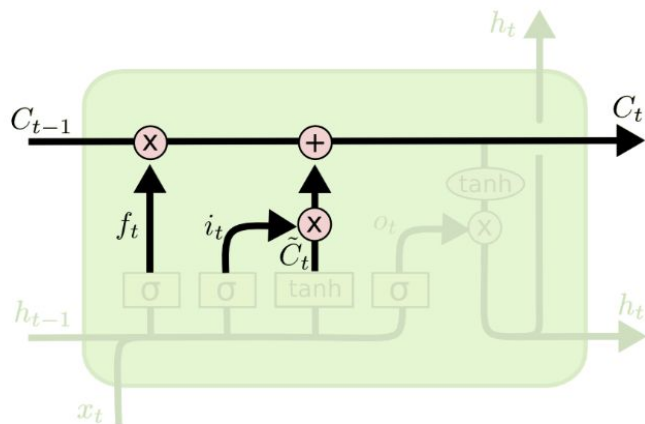
Input gate



$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

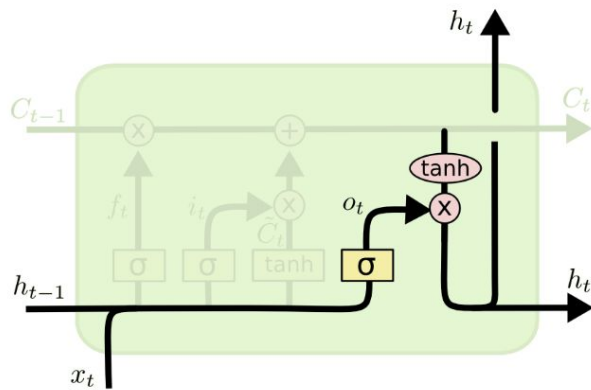
$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$$

Cell update



$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$

Output gate



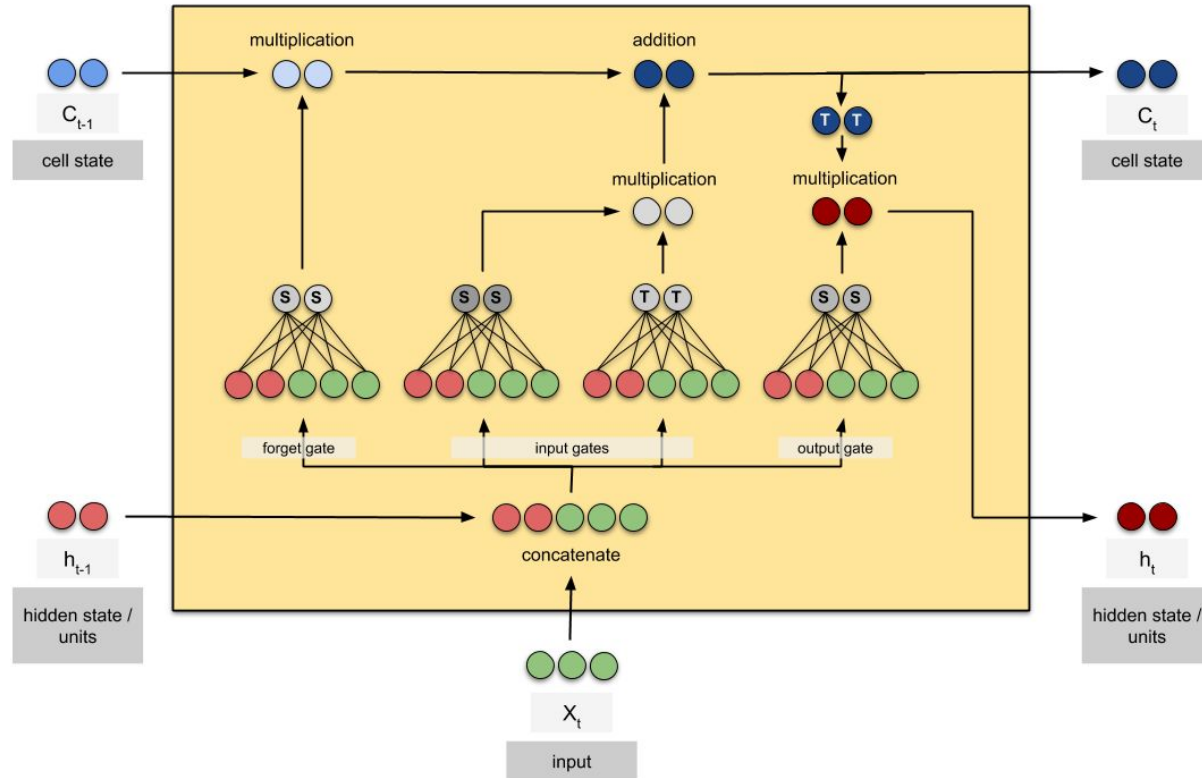
$$o_t = \sigma (W_o [h_{t-1}, x_t] + b_o)$$

$$h_t = o_t * \tanh (C_t)$$

Useful blog from Colah

<https://colah.github.io/posts/2015-08-Understanding-LSTMs/>

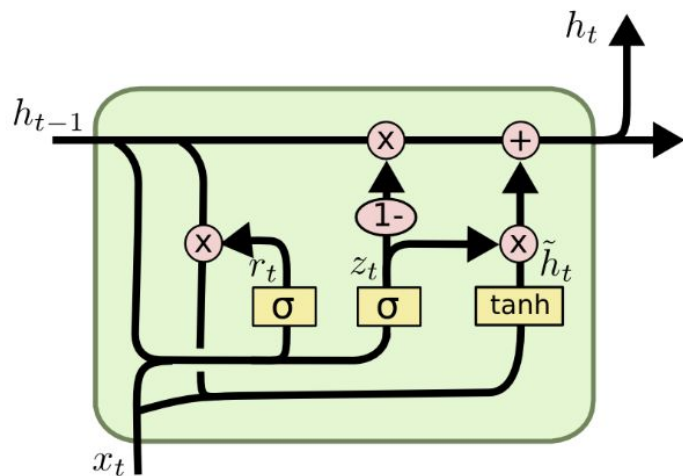
One LSTM cell



LSTM

- Step 1: Calculate forget (f_t), input (i_t) and output gates (o_t)
- Step 2: Calculate cell state update
- Step 3: Update cell state (C_t)
- Step 4: Calculate h_t

GRU



$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t])$$

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t])$$

$$\tilde{h}_t = \tanh(W \cdot [r_t * h_{t-1}, x_t])$$

$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t$$

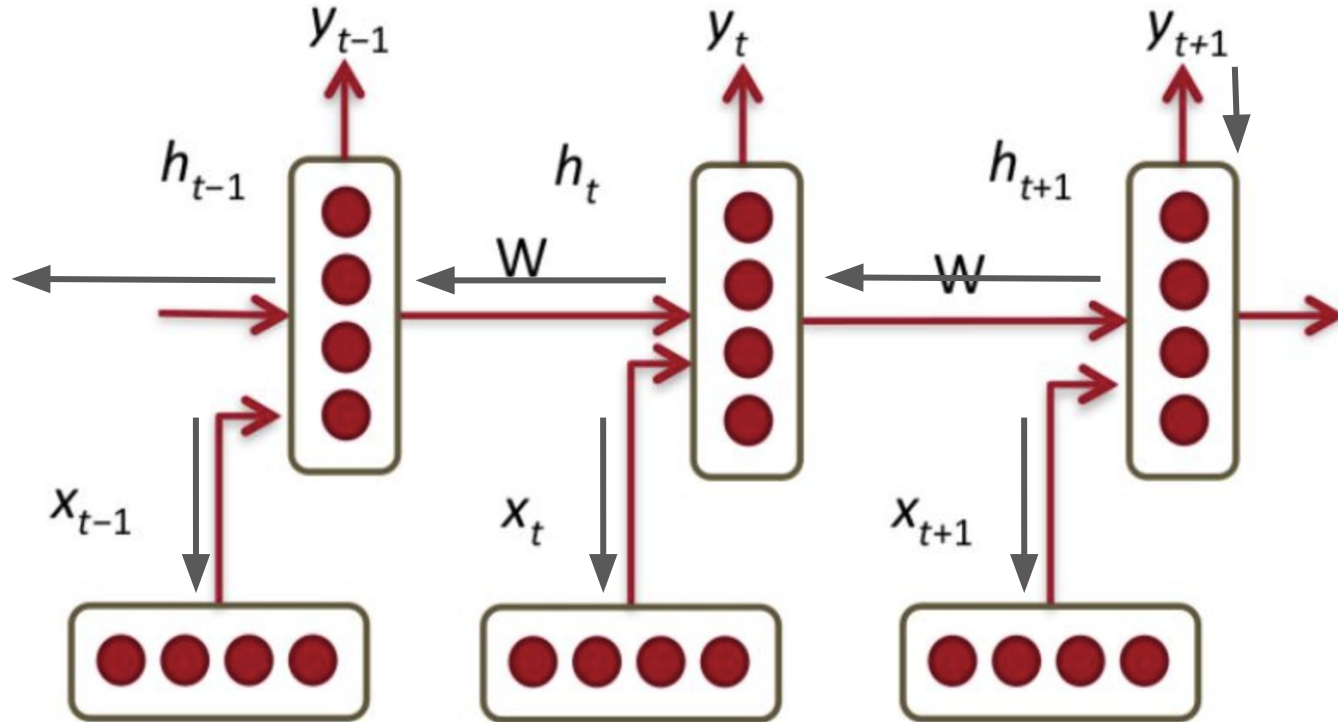
Gated RNNs

Properties:

- Can forget different amounts at each time step
- Much better at using long distance information

A bidirectional GRU is a good starting point for many sequence tagging tasks

Backpropagation through time



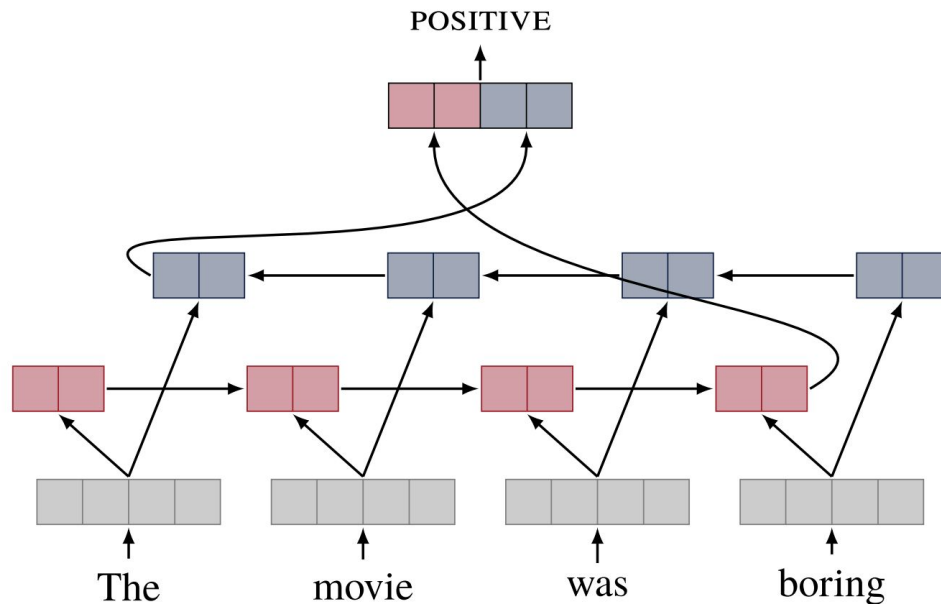
The vanishing gradient problem

- BPTT calculates gradients backward through time, it involves taking derivatives of the loss with respect to the model's parameters at each time step
- These derivatives are multiplied together as they are propagated backward
- Since gradients are multiplied together, if the gradients at each time step are less than 1, this multiplication leads to a compounding effect
 - As you go further back in time, the gradients become increasingly smaller
- The compounding effect causes the gradients for early time steps to become vanishingly small, approaching zero
 - When the gradients are too close to zero, they don't provide meaningful information for parameter updates
 - This makes it challenging for the RNN to learn long-term dependencies in the data

Recurrent architectures for related tasks

RNNs for text classification

- Last hidden state of the RNN represents the entire sentence.



Recurrent architectures for related tasks

What other tasks can RNNs handle?

Next to come

- Seq2seq models
 - Encoders and decoders
- Attention

Practice

Equations for LSTM

$$f_t = \text{sigmoid}(W_f x_t + U_f h_{t-1} + b_f)$$

$$i_t = \text{sigmoid}(W_i x_t + U_i h_{t-1} + b_i)$$

$$o_t = \text{sigmoid}(W_o x_t + U_o h_{t-1} + b_o)$$

- All the weights (W s and U s and b s) will be given. You need to calculate f , i and o .
- How to calculate sigmoid/tanh for a vector (because the argument for sigmoid in this case will be a vector)
 - $\text{sigmoid}([1, 2, 0]) = [\text{sigmoid}(1), \text{sigmoid}(2), \text{sigmoid}(0)]$
 - $\text{tanh}([1, 2, 0]) = [\text{tanh}(1), \text{tanh}(2), \text{tanh}(0)]$

Practice

How to calculate f_t for input $x_t = [1 \ 1]^T$

- Given: $W_f = [1 \ 1, \ 0 \ 1]$, $U_f = [0 \ 0, \ 2 \ 3]$, $h_{t-1} = [4 \ 5]^T$, $b_f = [0 \ 0]^T$
- Using this equation: $f_t = \text{sigmoid}(W_f x_t + U_f h_{t-1} + b_f)$
 - Multiply W_f with x_t (W_f has a shape of 2×2 , x_t has a shape of 2×1), output will be a 2×1 vector
 - Multiply U_f with h_{t-1} (U_f has a shape of 2×2 , h_{t-1} has a shape of 2×1), output will be a 2×1 vector
 - b_f already is a 2×1 vector
 - So, f_t will also be a 2×1 vector

Then calculate i_t and o_t and use these values to calculate the C_t and h_t

$$h_t = o_t * \tanh(C_t)$$

$$C_t = f_t * C_{t-1} + i_t * \hat{C}_t$$

$$\hat{C}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c)$$

P.S. * means element-wise multiplication

Practice

$$W_f X_t = [2 \ 1]^T, U_f h_{t-1} = [0 \ 23]^T \text{ bf } = [0 \ 0]^T, \text{ so } W_f x_t + U_f h_{t-1} + b = [2 \ 24]^T$$

$$f_t = \text{sigmoid}([2 \ 24]^T) = [\text{sigmoid}(2) \ \text{sigmoid}(24)]^T$$

$$C_t = f_t * C_{t-1} + i_t * \hat{C}_t$$

$$\text{If } C_{t-1} = [1 \ 5]^T$$

$$f_t * C_{t-1} = [\text{sigmoid}(2) \ \text{sigmoid}(6)]^T * [1 \ 5]^T$$

$$= [\text{sigmoid}(2)*1 \ \text{sigmoid}(24)*5]^T \text{ which is another } 2 \times 1 \text{ matrix}$$